

Typhoid Fever (MONDO:0005619): Comprehensive Disease Characteristics Report

Summary

Typhoid fever is a systemic, potentially life-threatening febrile illness caused by the **human-restricted** bacterium *Salmonella enterica* subspecies *enterica* serovar Typhi (*S. Typhi*). It is transmitted by the fecal–oral route through water and food contaminated by the excreta of acutely infected patients or, critically, of chronic gallbladder carriers who constitute a persistent human reservoir. Unlike the classical templates for Mendelian disorders, typhoid is an **infectious disease with no causal human gene**; host genetics act only as susceptibility modifiers. The Global Burden of Disease Study 2017 estimated **14.3 million (95% UI 12.5–16.3) cases** of typhoid and paratyphoid fevers in 2017 — a 44.6% decline from 1990 — with *S. Typhi* responsible for 76.3% of enteric-fever cases and a global case fatality of 0.95%. Burden is overwhelmingly concentrated in low- and middle-income countries (LMICs) of South Asia and sub-Saharan Africa, and incidence is highest in children under 15 years.

Mechanistically, the disease runs a well-characterized causal chain: ingested bacteria invade the small-intestinal epithelium and Peyer's-patch M cells via the SPI-1 type III secretion system, survive and replicate inside macrophages via the SPI-2 secretion system and the **Vi capsular polysaccharide** (which disturbs host autophagy and evades immune recognition), disseminate through the reticuloendothelial system (liver, spleen, bone marrow, gallbladder), and produce sustained bacteremia and a hyperinflammatory, endotoxin-driven fever. *S. Typhi* uniquely encodes a tripartite "**typhoid toxin**" (**CdtB + PltA + PltB**) that adds genotoxic and systemic effects. The most lethal complication is **ileal (intestinal) perforation** in weeks 2–3, carrying a pooled case-fatality of 15.4% among hospitalized cases.

Diagnosis still relies on blood culture (imperfect sensitivity), with rapid serologic tests offering only moderate accuracy. Treatment is increasingly constrained by **multidrug-resistant (MDR)** and **extensively drug-resistant (XDR)** strains — leaving azithromycin and carbapenems (meropenem) as the mainstays — while **typhoid conjugate vaccines (Vi-TT)** deliver ~80% efficacy with durable multi-year protection, and improved **water, sanitation and hygiene**

(WASH) remains the foundational preventive intervention. Chronic gallbladder carriage (2–5% of infections) not only sustains transmission but is quantitatively linked to gallbladder carcinoma (pooled OR $\approx$ 4.3).

1. Disease Information

Overview. Typhoid fever (enteric fever) is a systemic bacterial infection characterized by prolonged fever, bacteremia, and involvement of the reticuloendothelial system. It is caused by *Salmonella enterica* serovar Typhi, a Gram-negative, flagellated, facultatively intracellular bacillus that infects only humans. The closely related serovars Paratyphi A, B, and C cause a clinically similar but generally milder "paratyphoid fever"; together they constitute "enteric fever."

Key identifiers. - **MONDO:** MONDO:0005619 (typhoid fever) - **ICD-10:** A01.0 (Typhoid fever); **ICD-11:** 1A07 (Typhoid fever) - **MeSH:** D014435 (Typhoid Fever) - **SNOMED CT:** 4834000 (Typhoid fever) - **NCBI Taxonomy (pathogen):** *Salmonella enterica* subsp. *enterica* serovar Typhi — txid90370 - **OMIM/Orphanet:** Not a Mendelian disease; no OMIM disease entry. Host **susceptibility loci** have been mapped (see Section 4).

Synonyms / alternative names. Enteric fever (when grouped with paratyphoid), typhoid, "slow fever," historically "gastric fever" and "nervous fever."

Data provenance. The information in this report is derived almost entirely from **aggregated disease-level resources** — systematic reviews, meta-analyses, randomized controlled trials, GBD modeling, and controlled human infection (challenge) studies — rather than from individual EHR records.

2. Etiology

Primary cause (infectious). The sole causal agent is *S. Typhi* (**Finding F001**). Transmission is fecal–oral via contaminated water and food. The human-restricted nature of the pathogen means the reservoir is entirely human — acute cases and chronic carriers.

Environmental / behavioral risk factors. Inadequate **water, sanitation and hygiene (WASH)** is the dominant modifiable driver. A meta-analysis of 27 case-control studies quantified **limited hygiene (OR 2.26, 95% CrI 1.38–3.64)** and **untreated drinking water (OR 1.96)** as risk factors ([PMID: 37644449](#); **Finding F011**). Field studies add contaminated drinking water, poor sanitation, and **street-food consumption** as severity-associated exposures ([PMID: 42602150](#)). In Nairobi informal settlements, use of a **shared flush toilet** was independently associated with infection (aOR 2.42, 95% CI 1.69–3.47), as was age 5–16 years ([PMID: 42778887](#)). Additional host/behavioral factors include young age (school-age children), living in endemic/crowded urban slums, and — for the carrier state — gallstones, biliary abnormalities, and a cholesterol-rich diet.

Genetic risk factors (host). Typhoid is not a genetic disease, but host genetic susceptibility has been demonstrated. A controlled human challenge study genotyped volunteers and identified variants associated with enteric-fever susceptibility ([PMID: 35254093](#); **Finding F007**). Classical literature also implicates *HLA* class II alleles, *TLR* pathway variants, and the *CFTR* locus, though these are modifiers of modest effect rather than causal variants.

Protective factors. Vaccination (typhoid conjugate vaccine — Section 13) and improved household WASH are the principal protective factors; a Dhaka slum cohort (n = 98,087) found improved household WASH associated with a **38% reduction** in typhoid risk (adjusted HR 0.62, 95% CI 0.49–0.78) ([PMID: 37983081](#); **Finding F011**). Pre-existing immunity from prior exposure also protects.

Gene–environment interaction. The dominant interaction is between environmental exposure dose (WASH-mediated) and host immune competence; a persistent post-infection "convalescent" transcriptional signature may mark hosts genetically or temporarily unable to mount effective immunity, predisposing to relapse or the carrier state ([PMID: 20018727](#); **Finding F007**).

3. Phenotypes

Typhoid presents with an **insidious onset** of sustained fever over the first week, progressing over 2–3 weeks if untreated. In a retrospective series of 305 confirmed cases, **fever was universal (100%)**, followed by abdominal pain (62.0%), loss of appetite (43.3%), headache (40.3%), diarrhea (34.8%), and constipation (24.3%); hepatomegaly (30.1%) and splenomegaly (22.3%) were common ([PMID: 41246786](#)).

Phenotype	Type	HPO term	Approx. frequency	Notes
Prolonged / stepwise fever	Symptom	HP:0001945 (Fever)	~100%	Hallmark; rises over week 1
Abdominal pain	Symptom	HP:0002027	~62%	Diffuse, RLQ tenderness
Anorexia / loss of appetite	Symptom	HP:0002039	~43%	Common
Headache	Symptom	HP:0002315	~40%	Early
Diarrhea	Symptom	HP:0002014	~35%	More common in children
Constipation	Symptom	HP:0002019	~24%	Classic in adults
Hepatomegaly	Clinical sign	HP:0002240	~30%	Reticuloendothelial involvement
Splenomegaly	Clinical sign	HP:0001744	~22%	Reticuloendothelial involvement
Relative bradycardia (Faget sign)	Clinical sign	HP:0001662 (Bradycardia)	Variable	Classic but inconsistent
Rose spots (blanching macules)	Physical manifestation	HP:0011276 (Abnormality of skin morphology)	Variable	Trunk, ~week 2
Eosinopenia / leukopenia	Laboratory abnormality	HP:0001882 (Leukopenia)	Common	Supportive lab clue
Encephalopathy / "typhoid state"	Behavioral/ neuro	HP:0001298 (Encephalopathy)	Severe cases	Delirium, apathy
Intestinal perforation	Physical manifestation	HP:0031368 (Intestinal perforation)	severe/late	Week 2–3, lethal

Onset: subacute/insidious (days). **Severity:** variable — from mild self-limited febrile illness to fulminant disease with perforation, shock, and death. **Progression:** progressive over weeks if untreated, then resolving (with treatment) or complicated. **Quality-of-life impact:** substantial acute morbidity (mean hospital stay for perforation 18.4 days; **Finding F006**) and major economic burden — in hospitalized Kenyan children, median societal cost per admission US\$96.95, with up to ~24–27% of poorest households facing catastrophic health expenditure ([PMID: 42580776](#)).

4. Genetic/Molecular Information

Causal human genes: None. Typhoid is an infectious disease; there is no causal germline mutation.

Host susceptibility loci: Controlled human infection modeling has identified genetic variants associated with enteric-fever susceptibility ([PMID: 35254093](#)). Candidate genes in the broader literature include *HLA-DRB1/DQB1*, *TLR4/TLR5*, *SLC11A1* (NRAMP1), and *CFTR* (the CFTR protein has been proposed as an intestinal entry receptor for *S. Typhi*). These are **modifier/susceptibility** loci, not Mendelian causes.

Pathogen genetic determinants (the relevant "molecular" biology). Virulence is encoded on *Salmonella* pathogenicity islands (SPIs) and the *viaB* locus: - **SPI-1** — invasion-associated type III secretion system (T3SS-1). - **SPI-2** — intracellular survival T3SS-2. - **SPI-7 / *viaB* locus** — encodes the **Vi capsular polysaccharide** (*tviA*–*E*, *vexA*–*E*). - **Typhoid toxin operon** — *cdtB*, *pltA*, *pltB* (**Finding F002**).

Antimicrobial-resistance genetics (the clinically decisive molecular story). Resistance is driven by **plasmid acquisition and clonal spread of the H58 lineage**. Emerging cephalosporin resistance in India results from *bla* (e.g., *bla*CTX-M-15) acquisition via plasmids from other bacteria ([PMID: 40208005](#); **Finding F005**). Fluoroquinolone resistance arises from *gyrA/parC* QRDR mutations. XDR strains combine resistance to first-line drugs, fluoroquinolones, and third-generation cephalosporins ([PMID: 41550837](#)).

Epigenetic / chromosomal abnormalities: Not applicable to the human host in the Mendelian sense. (Bacterial DNA methylation regulates virulence gene expression but is outside the human-disease-gene framework.)

5. Environmental Information

Environmental factors. Contaminated water supplies and inadequate sewage/sanitation infrastructure are the principal environmental determinants; fecal contamination of drinking water is the classic transmission route. Wastewater surveillance detects *S. Typhi* in the community and correlates with clinical incidence — each 10-fold increase in typhoid incidence gave 2.43× higher odds of *S. Typhi* wastewater detection ([PMID: 42419341](#)).

Lifestyle factors. Street-food consumption, unsafe water handling and storage, and shared sanitation facilities increase risk ([PMID: 42602150](#); [PMID: 42778887](#)). A cholesterol-rich diet favors gallbladder carriage ([PMID: 39636114](#)).

Infectious agent. *Salmonella enterica* subsp. *enterica* serovar Typhi (NCBI Taxon 90370) — the necessary and sufficient cause. CHEBI-relevant chemical entities: lipopolysaccharide/endotoxin (CHEBI:16412), cholesterol (CHEBI:16113).

6. Mechanism / Pathophysiology

Ordered causal chain

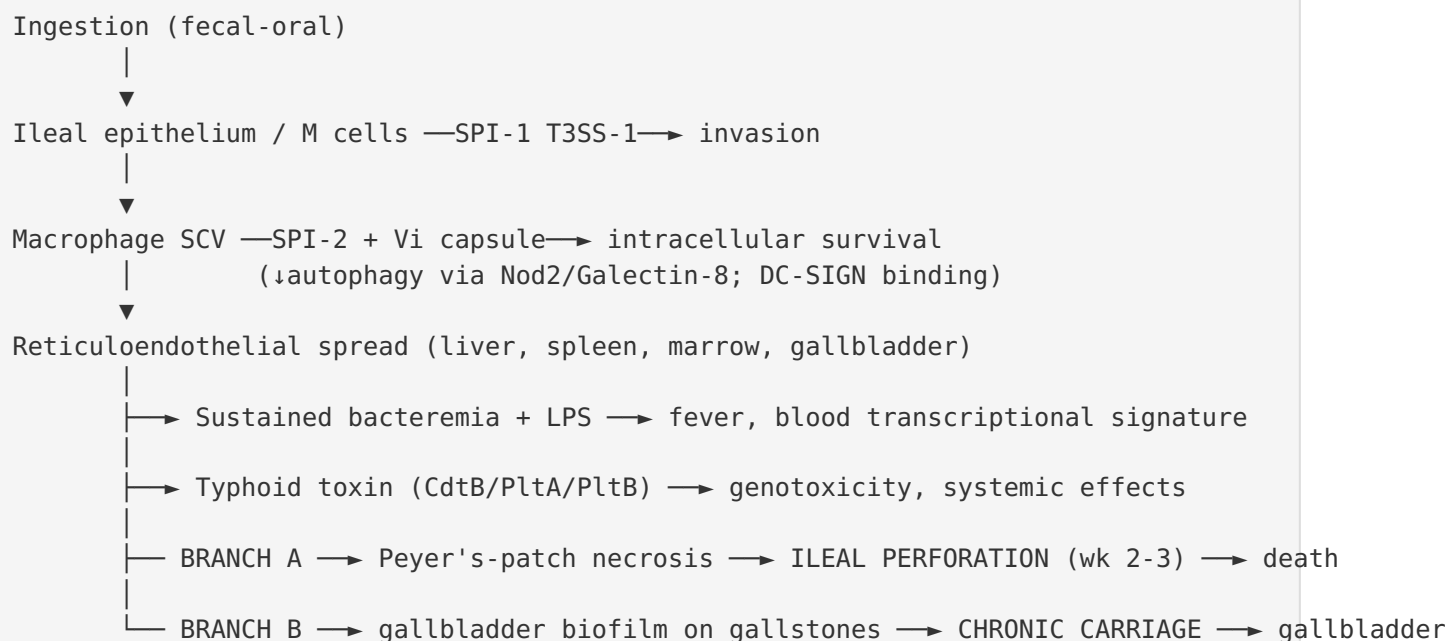
1. **Ingestion** of *S. Typhi* in fecally contaminated water/food → bacteria survive gastric acid and reach the small intestine. (*demonstrated*)
2. **SPI-1 T3SS-1-mediated invasion** of intestinal epithelium and **Peyer's-patch M cells** in the terminal ileum → bacterial translocation across the mucosa. (*demonstrated*)
3. **Uptake by macrophages/dendritic cells**; *S. Typhi* resides in the **Salmonella-containing vacuole (SCV)** and uses **SPI-2 T3SS-2** and the **Vi capsule** to survive intracellularly. The Vi capsule **decreases macrophage autophagy** by down-regulating Nod2 and Galectin-8 (↑ LC3-II, ↓ p62 in Vi mutants) → enhanced intracellular survival ([PMID: 39732413](#); **Finding F010**). (*demonstrated*)
4. **Immune evasion / modulation**: Vi paradoxically binds the human C-type lectin **DC-SIGN** to modulate phagocytosis ([PMID: 36286551](#)); relative to *S. Typhimurium*, *S. Typhi* induces **greater SPI-1-dependent inflammasome activation** (caspase-1, IL-1β, pyroptosis) in monocyte-derived macrophages → hyperinflammatory bacteremic state ([PMID: 32387390](#); **Finding F010**). (*demonstrated in vitro*)

5. **Reticuloendothelial dissemination:** infected macrophages carry bacteria to mesenteric lymph nodes, then via lymphatics/blood to **liver, spleen, bone marrow, and gallbladder** → primary/secondary bacteremia. (*inferred from classical pathology + demonstrated tropism*)
6. **Sustained bacteremia + endotoxin (LPS)–driven cytokine response** → prolonged stepwise fever, and a reproducible **peripheral-blood transcriptional signature** during acute disease ([PMID: 20018727](#); **Finding F007**). (*demonstrated*)
7. **Typhoid toxin action:** *S. Typhi* expresses CdtB within the SCV, secreted in outer-membrane vesicles; PltA/PltB deliver the genotoxic **CdtB** to target cells via retrograde Golgi transport → DNA damage, cell-cycle arrest, and contribution to systemic symptoms ([PMID: 18191792](#), [PMID: 23869968](#); **Finding F002**). (*demonstrated in vitro/in vivo*)

Branch A — acute severe disease: Hyperplasia and necrosis of Peyer's-patch lymphoid tissue in the terminal ileum → mucosal ulceration → **ileal perforation (weeks 2–3)** → peritonitis, sepsis, death (pooled CFR 15.4%; **Finding F006**).

Branch B — chronic carriage: Gallbladder colonization, favored by **gallstones and cholesterol-rich diet**, with **CsgD-regulated curli/cellulose biofilm** formation → asymptomatic intermittent shedding (2–5% of infections), antibiotic tolerance, transmission, and long-term **gallbladder carcinoma** risk ([PMID: 41335321](#), [PMID: 39720794](#), [PMID: 24612190](#); **Findings F003, F008**).

Mechanism map (ASCII)



Ontology suggestions. GO biological processes: GO:0009405 (pathogenesis), GO:0052167 (modulation of host immune response), GO:0006909 (phagocytosis), GO:0016236 (macroautophagy), GO:0002526 (acute inflammatory response). Cell types (CL): CL:0000235 (macrophage), CL:0000451 (dendritic cell), CL:0000236 (B cell), CL:0000084 (T cell), CL:0002270 (M cell / microfold cell). Subcellular (GO CC): GO:0005764 (lysosome), GO:0005794 (Golgi). Chemical entities (CHEBI): CHEBI:16412 (LPS), CHEBI:16113 (cholesterol).

7. Anatomical Structures Affected

- **Primary organs:** terminal **ileum / small intestine** (UBERON:0002116; Peyer's patches UBERON:0011156), **mesenteric lymph nodes** (UBERON:0002509).
 - **Reticuloendothelial system:** **liver** (UBERON:0002107; hepatomegaly ~30%), **spleen** (UBERON:0002106; splenomegaly ~22%), **bone marrow** (UBERON:0002371), **gallbladder** (UBERON:0002110 — carrier niche).
 - **Secondary/complication sites:** **peritoneum** (perforation → peritonitis), **CNS** (typhoid encephalopathy), skin (rose spots), and — in chronic carriers — gallbladder neoplasia.
 - **Body systems:** digestive, lymphatic/reticuloendothelial, hematopoietic, and (severe cases) nervous and cardiovascular systems.
 - **Tissue/cell level:** intestinal epithelium and **M cells** (CL:0002270); **macrophages** (CL:0000235) as the primary intracellular niche and key determinant of disease progression ([PMID: 40095029](#)); lymphocytes (T and B cells) recruited to the gallbladder in carriage (mouse model, [PMID: 31575775](#)).
 - **Subcellular:** the **Salmonella-containing vacuole**, lysosome, and Golgi (typhoid-toxin retrograde transport).
 - **Lateralization:** not applicable (systemic disease); ileal lesions are segmental, antimesenteric.
-

8. Temporal Development

- **Onset:** insidious/subacute; incubation typically **6–30 days** (usually ~1–2 weeks), dose-dependent.
- **Classic weekly progression (untreated):**

- Week 1 — rising stepwise fever, headache, malaise, relative bradycardia.
- Week 2 — sustained high fever, abdominal pain, rose spots, hepatosplenomegaly, "typhoid state" (apathy/delirium).
- Week 3 — risk of **ileal perforation and intestinal hemorrhage**, peritonitis, shock. Perforation cases nearly all present in the second week of infection ([PMID: 34652510](#)).
- Week 4+ — gradual resolution if survived.
- **Duration:** typically **self-limited to weeks** with treatment; relapse in ~5–10% after apparent recovery. **Chronic carriage** (>12 months of shedding) develops in 2–5% — a lifelong reservoir state (**Finding F003**).
- **Critical intervention window:** early antimicrobial therapy (first week) prevents most complications; the peri-perforation window (week 2–3) is the critical period for surgical intervention.

9. Inheritance and Population

Inheritance: Not applicable (infectious, non-heritable). Host susceptibility is **multifactorial/polygenic** with modest-effect loci.

Epidemiology (Findings F001, F009).

Metric	Estimate	Source
Global cases (enteric fever), 2017	14.3 million (95% UI 12.5–16.3)	PMID: 30792131
Trend 1990→2017	44.6% decline from 25.9 million	PMID: 30792131
Age-standardized incidence, 2017	197.8 / 100,000 person-years (↓ 54.9%)	PMID: 30792131
<i>S. Typhi</i> share of enteric fever	76.3% (71.8–80.5)	PMID: 30792131
Global case fatality, 2017	0.95% (0.54–1.53)	PMID: 30792131
WHO EMR annual cases	5.57–9.23 million; 46,200–163,000 deaths	PMID: 42413480
Sentinel-site median incidence	140 / 100,000 person-years	PMID: 42142522

Geographic distribution: endemic across **South Asia** (India, Pakistan, Bangladesh, Nepal), **sub-Saharan Africa**, and Southeast Asia; sporadic/travel-associated in high-income countries. Typhoid shows strong **spatial-temporal clustering** — incidence-rate ratio **4.9** in the innermost ring around index cases within 28 days, supporting targeted "ring" vaccination ([PMID: 38913735](#); **Finding F011**).

Age/sex: Incidence highest in **children <15 years** (school-age); a slight male predominance is often reported (e.g., male-to-female ~1.4:1 in a Lahore cohort, [PMID: 41246786](#)). **Chronic carriage** is more common in older adults, women, and those with gallstones.

10. Diagnostics

Reference standard: Blood culture (and, more sensitively, bone marrow culture). Blood culture positivity is imperfect (e.g., 61–78% in various cohorts: [PMID: 31884434](#), [PMID: 41246786](#)), reduced further by prior antibiotics.

Laboratory clues: leukopenia, eosinopenia, anemia; elevated liver transaminases. Anemia is prominent in perforation cases (62.5% in a Burkina Faso series, [PMID: 28406420](#)).

Rapid diagnostic tests (RDTs) / serology (moderate accuracy): A Cochrane review of 37 studies (5,080 participants) found **TUBEX** sensitivity 78% / specificity 87%; **Typhidot** sensitivity ~78–84% / specificity ~77–79%; **Test-It Typhoid (KIT)** sensitivity 69% / specificity 90% — all only moderately accurate ([PMID: 28545155](#)). The **Widal test** performs poorly and is not recommended alone. A Bayesian latent-class network analysis found IgM-based tests outperform IgG counterparts, with lateral-flow IgG and Reverse Passive Hemagglutination performing best in South Asian pediatric populations ([PMID: 31067228](#)).

Emerging biomarker assays: Plasma **IgA responses to HlyE + LPS** distinguished acute typhoid from other bacteremic illnesses (AUC 0.95; sensitivity 90%, specificity 92%) ([PMID: 30020426](#)) — a promising next-generation diagnostic.

Molecular: Multiplex qPCR targeting *ttr*, *staG*, *tviB* (used in wastewater surveillance, [PMID: 42419341](#)); gene markers *invA* (genus), *phsB* (H₂S), and *tviA* (S. Typhi-specific) for identification ([PMID: 42584516](#)).

Imaging: Erect abdominal radiograph showing **pneumoperitoneum** (86.7% of pediatric perforation cases, [PMID: 34652510](#)) for the perforation complication; ultrasound for hepatosplenomegaly and gallstones.

Genetic/omics diagnostics: Not applicable for host diagnosis. Pathogen whole-genome sequencing is used for AMR surveillance and outbreak tracing.

Differential diagnosis: malaria, dengue, rickettsial disease, leptospirosis, brucellosis, amebic liver abscess, and other causes of prolonged fever in endemic areas.

11. Outcome/Prognosis

- **Mortality:** With prompt appropriate antibiotics, case-fatality is low (global ~0.95%, [PMID: 30792131](#)). Untreated, historical mortality reached 10–30%.
- **Ileal perforation (principal lethal complication; Finding F006):** Systematic review of 42 reports — **4,626 cases, 706 deaths, pooled CFR 15.4% (95% CI 13.0–17.8%)**, mean hospital stay 18.4 days ([PMID: 24743649](#)). Single-center CFRs range 17–30% ([PMID: 28406420](#), [PMID: 24858189](#)); pediatric perforation mortality ~26.7%, reaching **100% with three or more perforations** ([PMID: 34652510](#), [PMID: 24858189](#)).
- **Prognostic factors:** number of perforations, severity of peritonitis, delay to surgery (>2 h operative time), tachycardia/tachypnea at presentation, and serum potassium (dominant predictor of prolonged stay in a pediatric ML model, [PMID: 42048896](#)).
- **Other complications:** GI hemorrhage, typhoid encephalopathy, myocarditis, relapse (~5–10%), chronic carriage (2–5%), and long-term **gallbladder carcinoma** (pooled OR 4.28; **Finding F008**).
- **Recovery:** most treated patients recover fully; carriage and gallbladder-cancer risk are the key long-term sequelae.

12. Treatment

Pharmacotherapy (empiric, guided by local resistance).

Drug / class	Role	NCIT (suggested)	Notes
Ceftriaxone (3rd-gen cephalosporin)	First-line where susceptible	C1096	Failing in XDR strains
Azithromycin (macrolide)	Uncomplicated & XDR	C1174	96–100% susceptible in recent cohorts
Meropenem / carbapenems	Severe / XDR	C61796	98–100% susceptible; mainstay for XDR
Fluoroquinolones (ciprofloxacin)	Historically first-line	C2471	Widespread resistance now
Ampicillin, chloramphenicol, TMP-SMX	Older first-line	—	MDR resistance common

Resistance landscape (Finding F005): MDR = resistance to ampicillin, chloramphenicol, and TMP-SMX; XDR additionally resists fluoroquinolones and third-generation cephalosporins, first reported in Pakistan and spreading (largely H58 lineage). In one XDR cohort, **all isolates resisted ceftriaxone, ciprofloxacin, and first-line agents; meropenem and azithromycin remained 100% and 96.9% susceptible** (PMID: 41550837). By 2021, MDR and XDR proportions reached 14.7% and 43.4% in one Pakistani study (PMID: 38710290). Emerging cephalosporin resistance in India arises from plasmid-borne *bla* genes (PMID: 40208005). Combination regimens (e.g., meropenem + azithromycin) are used for refractory XDR pediatric cases (PMID: 38404085).

Surgical/interventional: For perforation — **exploratory laparotomy** with simple two-layer closure, ileal resection with anastomosis, or ileostomy, plus peritoneal lavage (NCIT: laparotomy C15320). Timing is critical; operative delay worsens outcome.

Supportive care: fluid/electrolyte resuscitation, antipyretics, nutrition, and correction of hypokalemia.

Experimental / emerging: Anti-virulence and quorum-sensing (LuxS/AI-2) inhibitors and quorum-quenching biotherapeutics are under investigation but remain preclinical (PMID: 42530739, PMID: 42250148); phage therapy and multi-omics/AI-guided target discovery are being explored.

Pharmacogenomics: No established host pharmacogenomic guidance specific to typhoid therapy.

13. Prevention

Primary prevention — vaccination (Finding F004). Typhoid conjugate vaccines (Vi-TT / Vi-CRM197) are the leading tool. In the Malawi phase-3 RCT (28,130 children), a single dose of Vi-TT gave **78.3% efficacy** with durable protection over ~4.3 years across all age groups including infants (PMID: 38281499). A meta-analysis of 4 trials (111,481 children) found the risk of blood-culture-confirmed typhoid after Vi-TT was **0.18 vs controls** (~80% efficacy) (PMID: 40788116); a test-negative design estimated **80.3% effectiveness** (PMID: 36442498). Protection may wane 3–5 years post-vaccination (especially in those vaccinated <2 years), and evidence increasingly supports a **booster dose at a longer interval** (PMID: 42556475). WHO recommends mass campaigns for children 9 months–15 years followed by routine infant introduction. Older Vi-polysaccharide and Ty21a live-oral vaccines exist but are less suited to young children.

Primary prevention — WASH (Finding F011). Safe drinking water, improved sanitation, and hand hygiene reduce transmission; improved household WASH cut typhoid risk 38% (PMID: 37983081). WASH and vaccination are complementary.

Secondary prevention: early diagnosis/treatment; **carrier detection and treatment** (and cholecystectomy for carriers with gallstones) removes reservoirs. **Wastewater surveillance** enables early outbreak detection and vaccine-impact monitoring (PMID: 42419341); **ring vaccination** around cases is supported by spatial clustering (PMID: 38913735).

Tertiary prevention: prompt surgical management of perforation; antimicrobial stewardship to preserve azithromycin/carbapenems.

Public health: sanitation infrastructure, safe food handling (street-food hygiene), health education. Community KAP gaps are large — in Lilongwe, only 8.2% had adequate typhoid knowledge and 16.7% were aware of TCV (PMID: 41783922), underscoring the need for education.

14. Other Species / Natural Disease

S. Typhi is **strictly human-restricted** — there is no natural animal reservoir or naturally occurring typhoid fever in other species. This host restriction reflects co-evolution of virulence factors (including host-adapted typhoid toxin; [PMID: 28993610](#)) with the human host.

- **Zoonotic potential:** none for *S. Typhi* (contrast with non-typhoidal *Salmonella*, which are zoonotic).
 - **Comparative biology:** related host-adapted serovars cause analogous systemic disease in animals (e.g., *S. Gallinarum* in poultry, *S. Dublin* in cattle, *S. Choleraesuis* in pigs), providing comparative models of *Salmonella* host adaptation. Pigs infected with *S. Typhimurium* show shedding-level-dependent cytokine/transcriptomic phenotypes relevant to carriage biology ([PMID: 24632525](#)).
 - **Orthologous virulence loci** (SPI-1, SPI-2) are conserved across *Salmonella* serovars.
-

15. Model Organisms

Because *S. Typhi* does not naturally infect other species, modeling relies on surrogates:

Model	System	Use / recapitulation	Limitation
Mouse — <i>S. Typhimurium</i>	Mammalian, in vivo	"Mouse typhoid": systemic salmonellosis modeling SPI-1/SPI-2 pathogenesis; bioluminescent imaging tracks infection/carriage (PMID: 39619287)	Uses a different serovar; lacks Vi capsule and typhoid toxin
Mouse chronic-carriage models	Mammalian, in vivo	Gallbladder carriage with Type-2 immune shift and T/B-cell recruitment to gallbladder (PMID: 31575775); cholesterol-rich diet + biofilm factors favor carriage (PMID: 39636114 , PMID: 41410426)	Serovar/host mismatch
Humanized mice	Mammalian, in vivo	Permit <i>S. Typhi</i> infection via human immune components	Incomplete reconstitution
Controlled human infection (challenge) model	Human, in vivo	Gold standard for susceptibility genetics, early transcriptomics, vaccine testing (PMID: 35254093 , PMID: 37725060)	Ethical/logistical constraints; controlled dose
Macrophage cell lines / MDMs	In vitro	Vi-capsule autophagy modulation, SPI-1 inflammasome activation (PMID: 39732413 , PMID: 32387390)	Reductionist
Rabbit immunization	Mammalian	Vaccine immunogenicity/functional antibody assays (PMID: 30018230)	Not a disease model

Databases: MGI (mouse), and *Salmonella* genomic resources (Enterobase, BV-BRC) for pathogen genetics.

Key Findings (with statistical evidence)

F001 — Human-restricted *S. Typhi* causes a large LMIC-concentrated burden

WHO EMR review: **5.57–9.23 million cases and 46,200–163,000 deaths annually**, mainly in LMICs; sentinel median incidence 140/100,000 py; highest in children <15 y. *"Annually, there are 5.57 to 9.23 million typhoid fever cases and 46,200 to 163,000 associated deaths, mainly in low- and middle-income countries."* (PMID: 42413480).

F002 — Unique tripartite "typhoid toxin"

CdtB (genotoxic subunit) + PltA/PltB (pertussis-toxin homologs) assemble into a holotoxin expressed within the SCV and secreted in outer-membrane vesicles; retrograde Golgi transport is required for DNA damage. *"PltA and PltB are required for the delivery of CdtB from an intracellular compartment to target cells via autocrine and paracrine pathways... this toxin, which we have named 'typhoid toxin'"* (PMID: 18191792); *"...expresses its CDT (named as Typhoid toxin) only in the Salmonella-containing vacuole (SCV) of infected cells"* (PMID: 23869968).

F003 — Chronic gallbladder carriage sustains transmission

2–5% of infections become chronic carriers; gallstones + CsgD-regulated biofilm enhance persistence and antibiotic resistance. *"Chronic carriers, accounting for 2-5% of infections, play a crucial role in disease transmission... often asymptomatic but intermittently shed bacteria"; "Gallstones are strongly associated with the chronic carrier state, providing a niche for bacterial biofilm formation..."* (PMID: 41335321).

F004 — Typhoid conjugate vaccine ~80% efficacious, durable

Meta-analysis of 4 trials (111,481 children): RR 0.18 after Vi-TT. *"Four trials with 111 481 children found the risk of blood culture-confirmed typhoid fever after Vi-TT to be 0.18 compared with nontyphoid vaccines."* (PMID: 40788116); field effectiveness 80.3% (PMID: 36442498); durable ≥ 4 years (PMID: 38281499).

F005 — MDR/XDR narrows treatment to azithromycin + carbapenems

"All isolates were resistant to ceftriaxone, ciprofloxacin, and first-line agents; meropenem and azithromycin remained 100% and 96.9% susceptible." (PMID: 41550837); *"New strains in India show resistance to third-generation cephalosporins due to plasmid acquisition from other bacteria"* (PMID: 40208005).

F006 — Ileal perforation is the major lethal complication

"a total of 4,626 hospitalized typhoid intestinal perforation cases and 706 deaths were recorded (CFR = 15.4%; 95% CI 13.0%-17.8%)" (PMID: 24743649).

F007 — Reproducible blood transcriptional signature, persistent convalescent signature

"typhoid fever induced a distinct and highly reproducible signature in the peripheral blood that changed during treatment and convalescence"; the persistent convalescent signature may mark hosts "more susceptible to reinfection, relapse, or the establishment of a carrier state." (PMID: 20018727); susceptibility variants via challenge genotyping (PMID: 35254093).

F008 — Chronic carriage linked to gallbladder carcinoma

"The overall OR for chronic *S. typhi* carrier state was 4.28 (95% CI: 1.84-9.96)." (PMID: 24612190).

F009 — GBD 2017 authoritative burden

"Globally, 14.3 million (95% UI 12.5-16.3) cases of typhoid and paratyphoid fevers occurred in 2017, a 44.6% decline from 25.9 million in 1990."; "*Salmonella enterica* serotype Typhi caused 76.3% of cases... global case fatality of 0.95% in 2017." (PMID: 30792131).

F010 — Vi capsule + SPI-1 drive macrophage survival and hyperinflammation

"Vi capsule of *S. Typhi* decreased autophagy of macrophages to increase its survival in host cells by decreasing the expression of Nod2 and Galectin-8." (PMID: 39732413); "*S. Typhi*, relative to its non-typhoidal counterpart, *S. Typhimurium*, induces greater SPI-1-dependent inflammasome activation in monocyte-derived macrophages" (PMID: 32387390).

F011 — WASH is a modifiable risk factor; typhoid clusters spatially

"Pooled estimates of limited hygiene (OR = 2.26, 95% CrI: 1.38 to 3.64), untreated water (OR = 1.96...)" (PMID: 37644449); "The IRR in this innermost cluster was 4.9" (PMID: 38913735); improved household WASH → 38% risk reduction (PMID: 37983081).

Mechanistic Model / Interpretation

The findings cohere into a single narrative in which **the pathogen's cellular tropism dictates both acute disease and the chronic reservoir**. *S. Typhi*'s defining evolutionary innovations — the **Vi capsule** and the **typhoid toxin** — are the molecular reasons it behaves so differently from non-typhoidal *Salmonella*: the Vi capsule permits an intracellular lifestyle (by suppressing autophagy and modulating DC-SIGN-mediated phagocytosis) that would be paradoxical for a normally anti-phagocytic surface structure, while the exaggerated SPI-1 inflammasome response explains the hyperinflammatory bacteremic state clinically seen as prolonged high fever (F010). The macrophage is therefore the pivotal cell (CL:0000235): its permissiveness allows reticuloendothelial dissemination, and the balance of its inflammatory response shapes whether disease is self-limited, fulminant (→ perforation, F006), or transitions to carriage.

The **carriage branch** (F003, F008) transforms an acute infection into a lifelong public-health problem: gallstone-associated biofilm creates an antibiotic-tolerant niche whose intermittent shedding sustains community transmission and whose chronic inflammation drives a >4-fold increase in gallbladder-cancer risk. This links an infectious disease to an oncologic outcome — a rare and important causal chain.

At the population level, the epidemiologic findings (F001, F009, F011) show the disease is **environmentally gated**: burden tracks WASH deficits and clusters spatially, so both WASH investment and geographically targeted vaccination are rational levers. The therapeutic findings (F005) and preventive findings (F004) frame the current crisis and its solution: as XDR strains erode the antibiotic armamentarium, the **conjugate vaccine** (durable ~80% efficacy) shifts control from cure toward prevention.

Evidence Base

PMID	Contribution	Relationship to findings
30792131	GBD 2017 burden	Anchors F009 (14.3M cases, 76.3% Typhi, CFR 0.95%)
42413480	WHO EMR burden	Supports F001
18191792 , 23869968	Typhoid toxin delivery/secretion	Define F002
39732413 , 32387390 , 36286551 , 40095029	Vi/SPI-1 macrophage mechanisms	Build F010
41335321 , 39720794 , 39636114	Carriage, biofilm, diet	Support F003
24612190	Carriage–gallbladder cancer meta-analysis	Quantifies F008 (OR 4.28)
24743649	Perforation CFR meta-analysis	Anchors F006 (CFR 15.4%)
40788116 , 38281499 , 36442498	Vi-TT efficacy/effectiveness	Establish F004
41550837 , 40208005 , 38710290	XDR resistance	Support F005
20018727 , 37725060 , 35254093	Host transcriptomics/genetics	Support F007
37644449 , 37983081 , 38913735	WASH & clustering	Support F011
28545155 , 31067228 , 30020426	Diagnostics	Section 10

Limitations and Knowledge Gaps

1. **No human causal genetics.** Because typhoid is infectious, the template's genetic sections (causal genes, ACMG variant classification, inheritance, penetrance) are largely not applicable; host susceptibility loci are of modest, incompletely mapped effect.
 2. **Burden uncertainty.** GBD estimates carry wide uncertainty intervals and depend on modeling; blood-culture under-ascertainment biases incidence downward. Serosurveys and wastewater surveillance are emerging but not standardized.
 3. **Mechanistic evidence is largely in vitro / surrogate-model.** Key macrophage mechanisms (Vi-autophagy, SPI-1 inflammasome) derive from cell lines; carriage biology relies on *S. Typhimurium* mouse models that lack Vi and typhoid toxin, limiting direct translation.
 4. **Diagnostics remain suboptimal.** Blood culture is insensitive and slow; RDTs are only moderately accurate; the promising IgA/HlyE assay is not yet widely deployed.
 5. **Vaccine durability.** Waning immunity 3–5 years post-TCV and optimal booster strategy are unresolved.
 6. **Paratyphoid gap.** *S. Paratyphi A* causes a growing share of enteric fever and is not covered by current TCVs.
-

Proposed Follow-up Experiments / Actions

1. **Deploy next-generation diagnostics:** validate and scale the IgA anti-HlyE/LPS assay ([PMID: 30020426](#)) and standardize wastewater qPCR surveillance for real-time transmission monitoring and vaccine-impact evaluation.
2. **Define human host-susceptibility architecture:** expand challenge-cohort GWAS ([PMID: 35254093](#)) and connect it to the persistent convalescent transcriptional signature to predict carriage/relapse risk.
3. **Test carriage-eradication strategies:** anti-biofilm/CsgD-targeted and quorum-quenching agents in gallbladder-carriage models, with the explicit endpoint of interrupting transmission and reducing gallbladder-cancer risk.
4. **Optimize vaccination policy:** trials of TCV booster timing (favoring longer intervals, [PMID: 42556475](#)) and evaluation of **ring vaccination** exploiting spatial clustering (IRR 4.9, [PMID: 38913735](#)).

5. **Preserve last-line antibiotics:** implement antimicrobial-stewardship programs and genomic AMR surveillance (H58 lineage, *bla* plasmid tracking) to protect azithromycin and carbapenems.
 6. **Advance bivalent/multivalent vaccines** covering *S. Paratyphi A* and iNTS to close the enteric-fever coverage gap ([PMID: 42566676](#)).
-

Report compiled from 11 confirmed findings and 59 reviewed papers across 5 investigation iterations. Evidence types span human clinical (RCTs, cohorts, meta-analyses), controlled human infection, model organism (mouse), and in vitro mechanistic studies.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery